

Equipe

Piège harmonique anisotrope

Pièges optiques ou magnétiques

Confinement transversal fort, anisotrope

Gerbier (2004)

Réseaux optiques 2D

Lattice de faisceaux croisés

Ensemble de tubes 1D

Paredes (2004),
Kinoshita (2004)

Pièges sur puce atomique

Fils micro-fabriqués,
RF-dressing

Tube unique fortement confiné

van Amerongen
(2008), Jacqmin (2011)

Résonance de confinement (CIR)

Toute géométrie confinée

Modification du couplage $g_1 D$

Olshanii (1998), Haller (2010)

One-dimensional quantum gases provide an ideal platform to study strong correlations and non-ergodic dynamics in out-of-equilibrium systems.

Despite major progress, understanding the local rapidity distribution remains a key challenge.

or Schematic view : from experimental system to theoretical description.

Packet d'onde semiclassique en collisions

graph (x,t) pour visualiser Δ

Integrable
description

1D Quantum Q
(Experiment)

Local
Measuremen

Rapidity
Distribution

Generalized
Gibbs Ensemble

Bridge between
experiment and theory